# Results with BCS theory

Now that we can solve Eqs. **??** and **??** let's see the results for some different materials and various temperatures. In particular i have tested Nb, Pb and Al.

For different materials we need the experimental values of thecritical temperature $T_c$, under which they become superconductors, and the zero temperature value of the enrgy gap $\Delta_0$. Actually for $\Delta_0$ we extract the value from the BCS relation

$$\frac{2\Delta_0}{K_B T_c} = \alpha, \tag{1}$$

where $\alpha$ is $\sim 3.53$ for weak-coupling materials, that is the limit where BCS theory and MB formulas holds, and is greater for strong-coupling materials such as Nb and Pb. In Tab. 1 below the $T_c$ and $\alpha$ values are showed.

Since $\Delta = \Delta(T)$ is a function of the temperature and is zero at $T \geq T_c$ i have extrapolate it in the BCS approximation as

$$\Delta = \Delta_0 \ \tanh\left(e^{\gamma}\sqrt{\frac{8}{7\xi(3)}}\ \sqrt{\frac{T_c}{T}-1}\right) \simeq \Delta_0 \ \tanh\left(1.74\ \sqrt{\frac{T_c}{T}-1}\right). \tag{2}$$

In the following we see some test for Nb, Pb and Al at different temperatures. In Fig. 1 i compared the real and imaginary conductivity for Nb, Pb and Al at $T = 0.8T_c$. In Fig. 2 we see conductivity of Nb for different temperatures. The last Fig. 3 i have plotted the conductivity of Niobium with the energy scale normalized by $2\Delta_0$ to check if the real part of $\sigma/\sigma_n$ correctly drops down at $\hbar\omega = 2\Delta_0$ when $T = 0$, and if increasing $T$ the gap moves to lower energies/frequencies.

Table 1: Superconducting materials experimental parameters.

| Material | $T_c$(K) | $\alpha$ |
|---|---|---|
| Nb | 9.3 | 3.65 |
| Pb | 7.2 | 3.95 |
| Al | 1.2 | 3.53 |

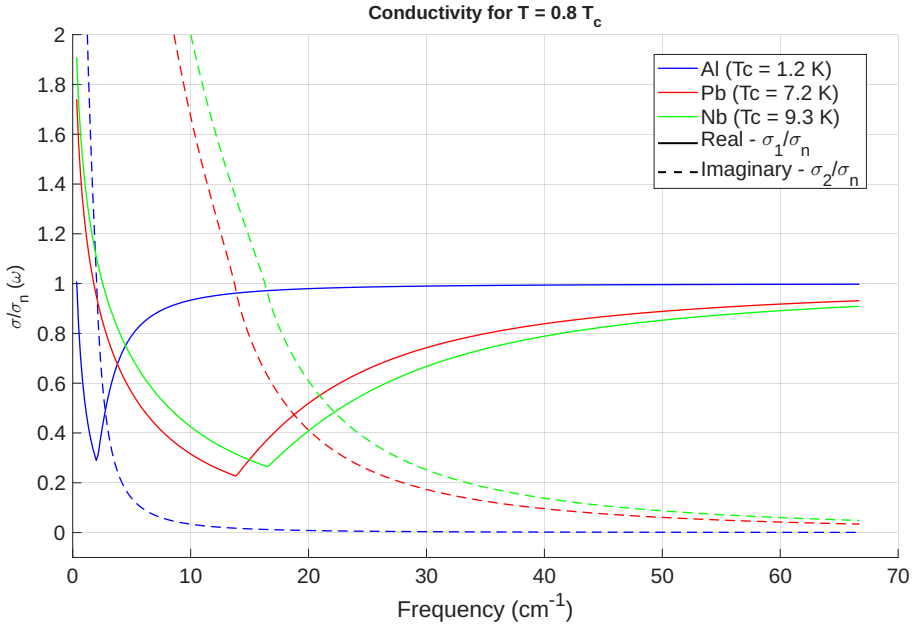



Figure 1: Complex conductivity for different materials.

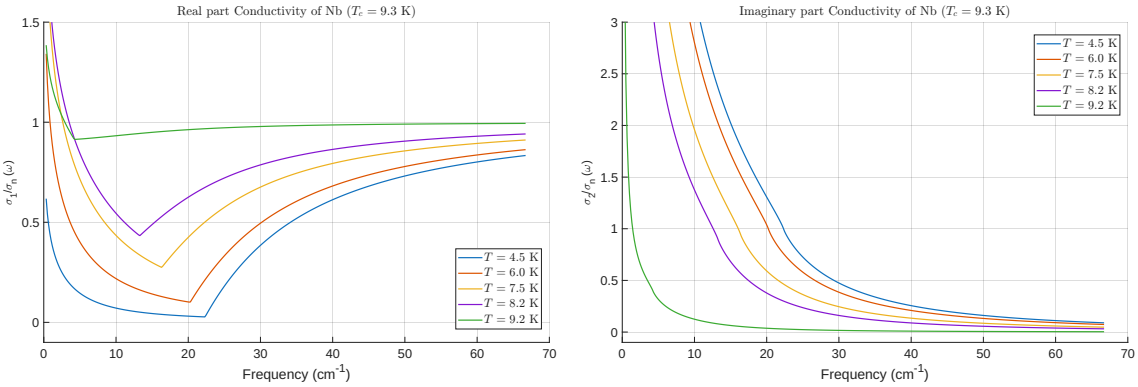



Figure 2: Real (sx) and imaginary (dx) conductivity for Nb.

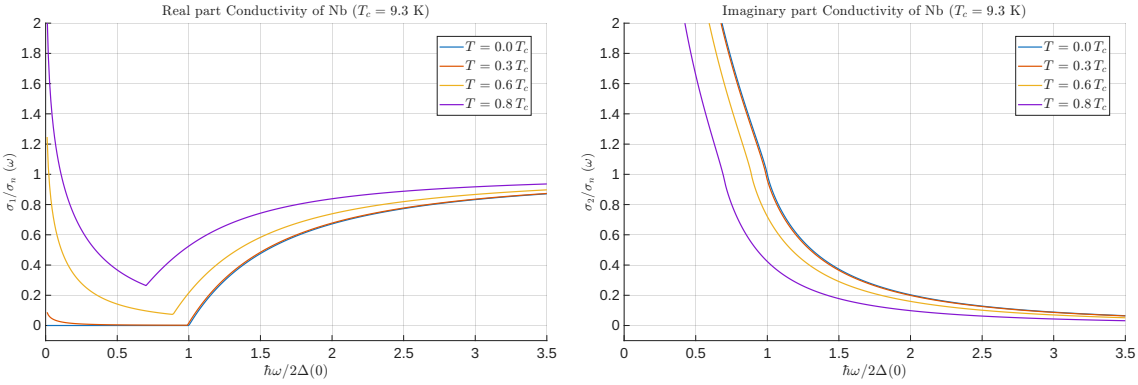



Figure 3: Real (sx) and imaginary (dx) conductivity for Nb in normalized units, to check the gap corresponds at $T = 0$.

## Comparison with experiments

To evaluate the performances of the code that i've written, i compare the numerical solutions with experimental data of Nb. To extract the data points from the figure i have used the online tool `WebPlotDigitizer`. The results are showed in the Fig.4 below with $T_c = 8.3$ K and $\alpha = 4.1$. We can see that the BCS understimates the value of the

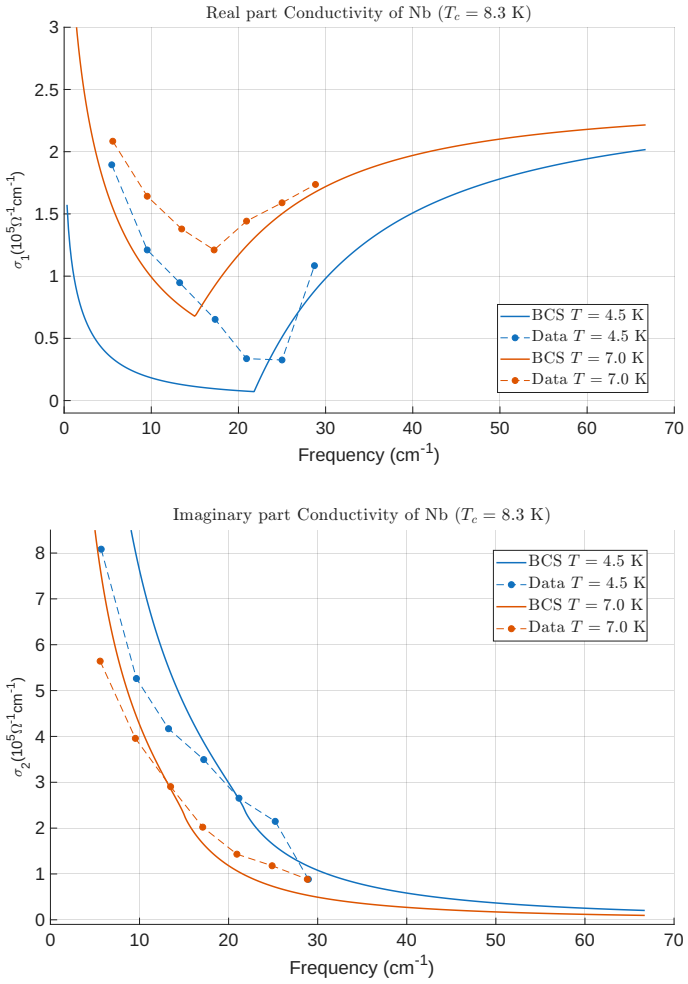



Figure 4: Real (sx) and imaginary (dx) conductivity for Nb compared with experimental data.

real conductivity $\sigma_1$, expecially for lower $T$, but we espected this for Niobium because is a strong-coupling superconducting material while BCS is a weak-coupling theory ($\alpha = 3.65$).

Another comparison that needs to be done is the one between the values of $\Delta$ at different temperatures extracted by the experimental data of conductivity, and the BCS mean field approximation used for the gap function in Eq. 2. The value of $2\Delta(T)$ is given by the frequency that corresponds to the minimum of the $\sigma_1$ at given temperature. This is showed in Fig. 5. Furthermore in Fig. 5 i have plotted the frequency at the minimum of $\sigma_1$ obtained from the numerical solution of Mattis-Bardeen as a counter check for the

code.

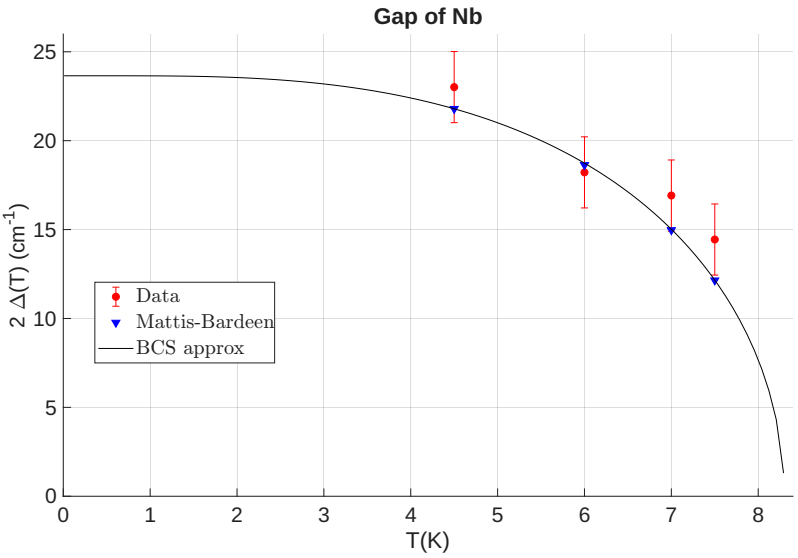



Figure 5: Comparison between BCS approximation of $\Delta(T)$ and the values of $\Delta$ extracted by the data. Errorbars are 2 $\text{cm}^{-1}$ from the paper [ref].